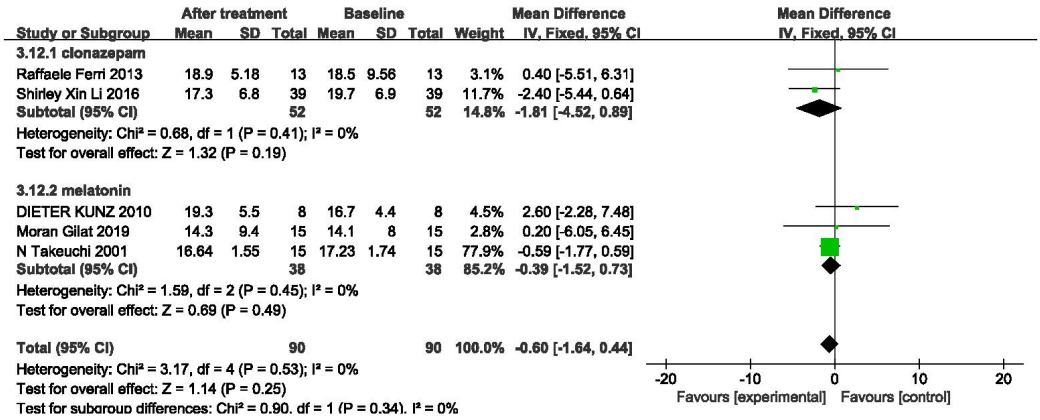
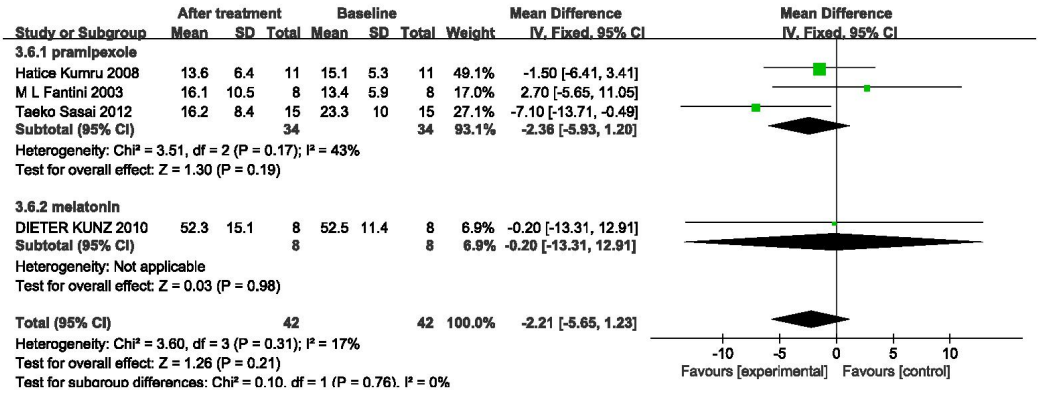
Data							
Total sleep time, min							
pramipexole	Authors & Year	Data					
		Baseline			After treatment		
		mean	SD	n	mean	SD	n
	M L Fantini 2003	390.5	43.5	8	390.4	65	8
	Hatice Kumru 2008	367.6	55.7	11	418.2	32.3	11
	Taeko Sasai 2012	402.8	63.4	15	409	88.6	15
clonazepam	Raffaele Ferri 2013	349.3	62.68	13	378.4	29.93	13
rotigotine	Yan Wang 2016	337.32	76.02	11	401.05	77.85	11
ramelteon	Takashi Nomura 2013	298	101.1	2	360	83.4	2
	Yuichi Esaki 2016	343	51.8	12	332.6	54.2	12
melatonin	DIETER KUNZ 2010	378.8	85.1	8	417	63.4	8
	Moran Gilat 2019	360.6	72.4	15	353.3	69	15
	N Takeuchi 2001	402.3	20.58	15	408.33	21.19	15
	D Kunz 1997	360	0	1	357	0	1
Sleep latency, min							
clonazepam	Authors & Year	Data					
		Before			After		
		mean	SD	n	mean	SD	n
	Shirley Xin Li 2016	18.9	14.1	39	23.6	29.2	39
	Raffaele Ferri 2013	24.6	25.67	13	18	13.94	13
rotigotine	Yan Wang 2016	33.64	38.11	11	17.5	10.08	11
REMs density, %							
pramipexole	Authors & Year	Data					
		Baseline			After treatment		
		mean	SD	n	mean	SD	n
	M L Fantini 2003	13.4	5.9	8	16.1	10.5	8
	Hatice Kumru 2008	15.1	5.3	11	13.6	6.4	11
	Taeko Sasai 2012	23.3	10	15	16.2	8.4	15
melatonin	DIETER KUNZ 2010	52.5	11.4	8	52.3	15.1	8
	D Kunz 1997	34	0	1	30	0	1
PLMS index							
pramipexole	Authors & Year	Data					
		Baseline			After treatment		
		mean	SD	n	mean	SD	n
	M L Fantini 2003	23.8	13.1	8	20.7	16	8
	Hatice Kumru 2008	13.2	27.1	11	0	0	11
	Taeko Sasai 2012	38.4	17.9	15	19.2	20.7	15
clonazepam	Raffaele Ferri 2013	18.3	25.84	13	20	20.99	13
rotigotine	Yan Wang 2016	24.86	11.37	11	18.81	11.46	11
Sleep efficiency (%)							
pramipexole	Authors & Year	Data					
		Baseline			After treatment		
		mean	SD	n	mean	SD	n
	Hatice Kumru 2008	76	10.3	11	85	6.5	11
	Taeko Sasai 2012	82.5	10.3	15	79.4	14.1	15
clonazepam	Shirley Xin Li 2016	73.3	11.9	39	75.9	11.1	39
	Raffaele Ferri 2013	78.3	14.61	13	85.4	6.41	13
rotigotine	Yan Wang 2016	65.27	12.31	11	73.98	12.7	11
melatonin	N Takeuchi 2001	82.87	2.48	15	85.26	2.5	15
	D Kunz 1997	77.9	0	1	76.3	0	1
	DIETER KUNZ 2010	76.4	16.1	8	86.4	11.8	8
Stage 1 (%)							
pramipexole	Authors & Year	Data					
		Baseline			After treatment		
		mean	SD	n	mean	SD	n
	Hatice Kumru 2008	22.1	17.7	11	22.3	15	11
	Taeko Sasai 2012	6.8	3.3	15	7.5	3.4	15
clonazepam	Shirley Xin Li 2016	13.4	5.4	39	12.7	5.2	39
	Raffaele Ferri 2013	9	7.11	13	6	2.74	13
rotigotine	Yan Wang 2016	29.9	9.99	11	36.27	10.81	11
Stage 2 (%)							
pramipexole	Authors & Year	Data					
		Baseline			After treatment		
		mean	SD	n	mean	SD	n
	Hatice Kumru 2008	46.8	15.2	11	51.8	11.9	11
	Taeko Sasai 2012	51.8	9.8	15	52.5	12.3	15
clonazepam	Shirley Xin Li 2016	65.2	7.7	39	69	8.4	39
	Raffaele Ferri 2013	43.3	7.57	13	47.9	8.49	13
rotigotine	Yan Wang 2016	45.79	13.44	11	46.12	15.6	11

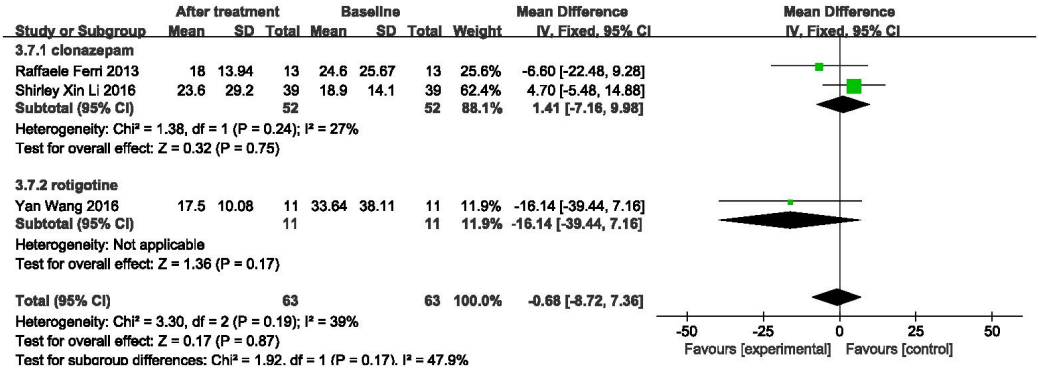
Stage 3–4 (%)							
	Authors & Year	Data					
		Baseline			After treatment		
pramipexole		mean	SD	n	mean	SD	n
	Hatice Kumru 2008	11.9	8	11	12.5	9.1	11
	Taeko Sasai 2012	5	5.5	15	3	2.8	15
rotigotine	Yan Wang 2016	8.75	7.17	11	6.3	6.19	11
Stage REM (%)							
	Authors & Year	Data					
		Baseline			After treatment		
pramipexole		mean	SD	n	mean	SD	n
	Hatice Kumru 2008	19.2	7.3	11	13.4	5.5	11
	Taeko Sasai 2012	11.3	4.9	15	14.7	6.8	15
ramelteon	Takashi Nomura 2013	8.4	0.7	2	11.9	4.9	2
	Yuichi Esaki 2016	15.9	6.7	12	16.9	9.3	12
SWS, %							
	Authors & Year	Data					
		Before			After		
clonazepam	Shirley Xin Li 2016	mean	SD	n	mean	SD	n
		1.1	2.1	39	1	3.4	39
	Raffaele Ferri 2013	12.5	8.34	13	18.5	6.73	13
melatonin	DIETER KUNZ 2010	10.2	6.8	8	7.6	5.7	8
REM, %							
	Authors & Year	Data					
		Before			After		
clonazepam	Shirley Xin Li 2016	mean	SD	n	mean	SD	n
		19.7	6.9	39	17.3	6.8	39
	Raffaele Ferri 2013	18.5	9.56	13	18.9	5.18	13
melatonin	DIETER KUNZ 2010	16.7	4.4	8	19.3	5.5	8
	Moran Gilat 2019	14.1	8	15	14.3	9.4	15
	N Takeuchi 2001	17.23	1.74	15	16.64	1.55	15
	D Kunz 1997	13.1	0	1	16.9	0	1
TIB, min							
	Authors & Year	Data					
		Before			After		
clonazepam	Raffaele Ferri 2013	mean	SD	n	mean	SD	n
		447.3	17.6	13	443.5	25.23	13
	Moran Gilat 2019	426.7	48.4	15	422.2	74.5	15
melatonin	N Takeuchi 2001	524.53	14.01	15	512.33	18.44	15
	D Kunz 1997	485	0	1	511	0	1
Sleep period time, min							
	Authors & Year	Data					
		Before			After		
clonazepam	Raffaele Ferri 2013	mean	SD	n	mean	SD	n
		418.9	30.57	13	415.6	30.27	13
melatonin	DIETER KUNZ 2010	493	39.5	8	479.9	17	8
	D Kunz 1997	476	0	1	480	0	1
ESS							
	Authors & Year	Data					
		Baseline			End of treatment		
rotigotine	Yan Wang 2016	mean	SD	n	mean	SD	n
		5	2.49	11	2.49	3.79	11
melatonin	Jin-Sun Jun 2019	9.6	3.5	9	8.2	5	9



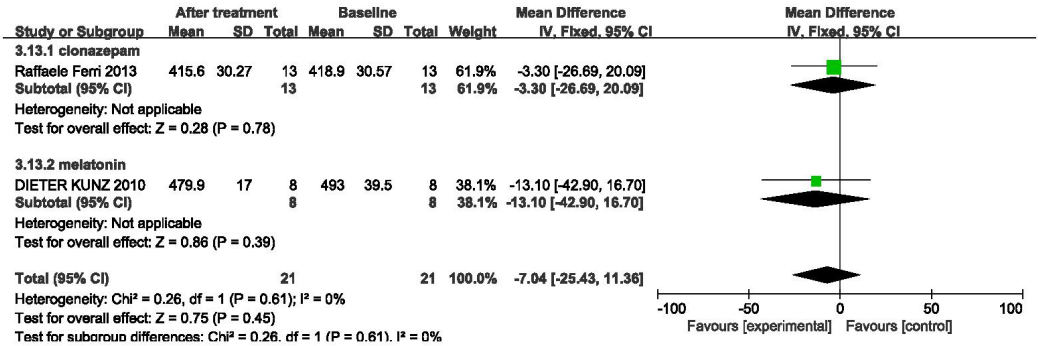
Supplementary figure 1: Forest plot of the meta-analysis showing the effect of different drugs on REM



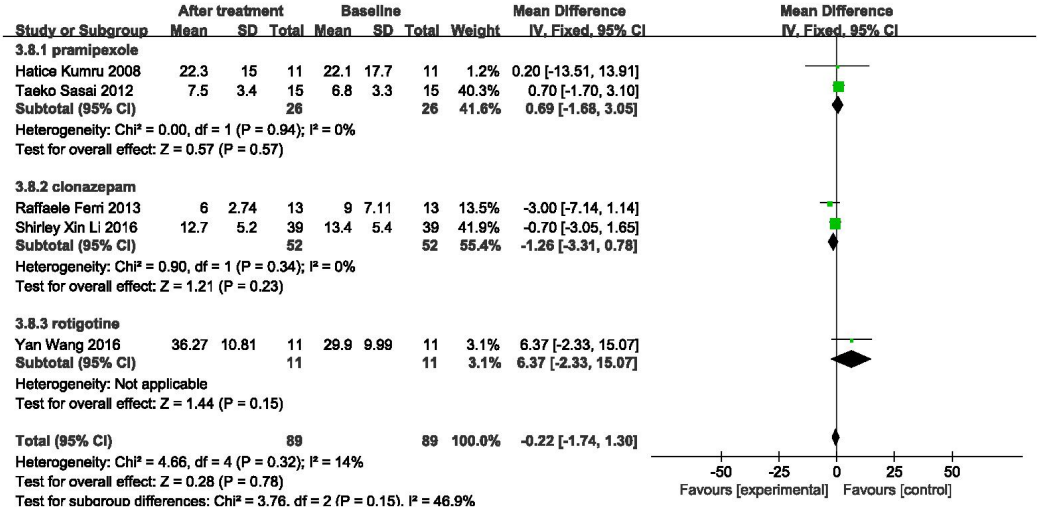
Supplementary figure 2: Forest plot of the meta-analysis showing the effect of different drugs on REMs density



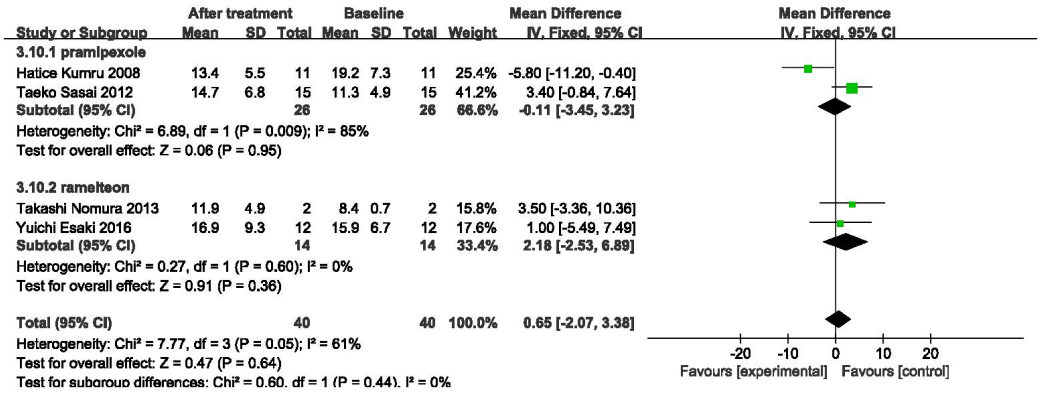
Supplementary figure 3: Forest plot of the meta-analysis showing the effect of different drugs on Sleep latency



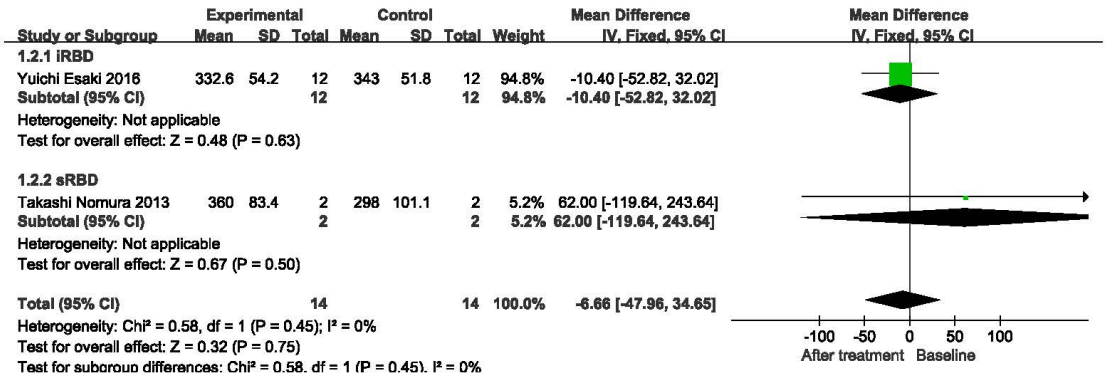
Supplementary figure 4: Forest plot of the meta-analysis showing the effect of different drugs on Sleep period time



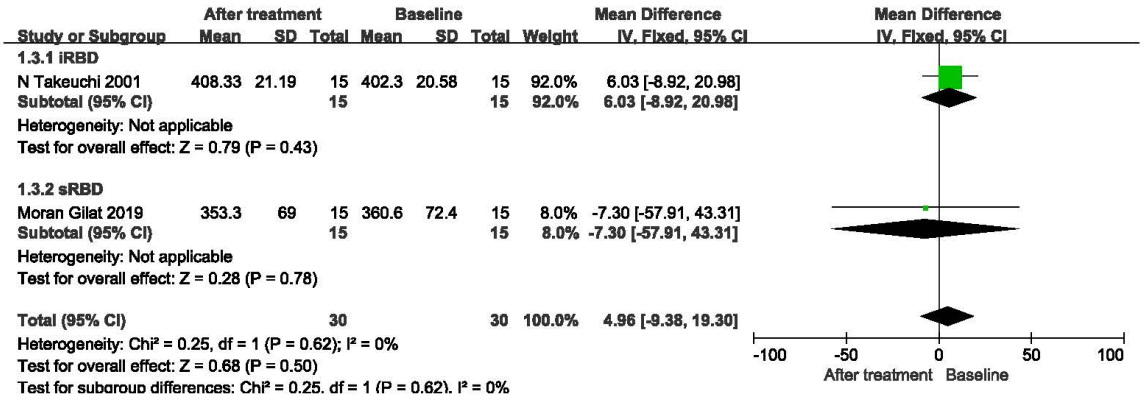
Supplementary figure 5: Forest plot of the meta-analysis showing the effect of different drugs on Stage 1 sleep



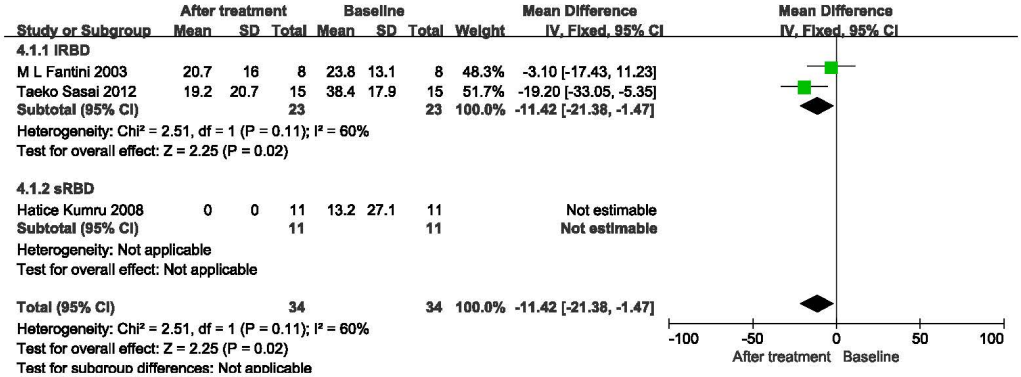
Supplementary figure 6: Forest plot of the meta-analysis showing the effect of different drugs on stage REM sleep



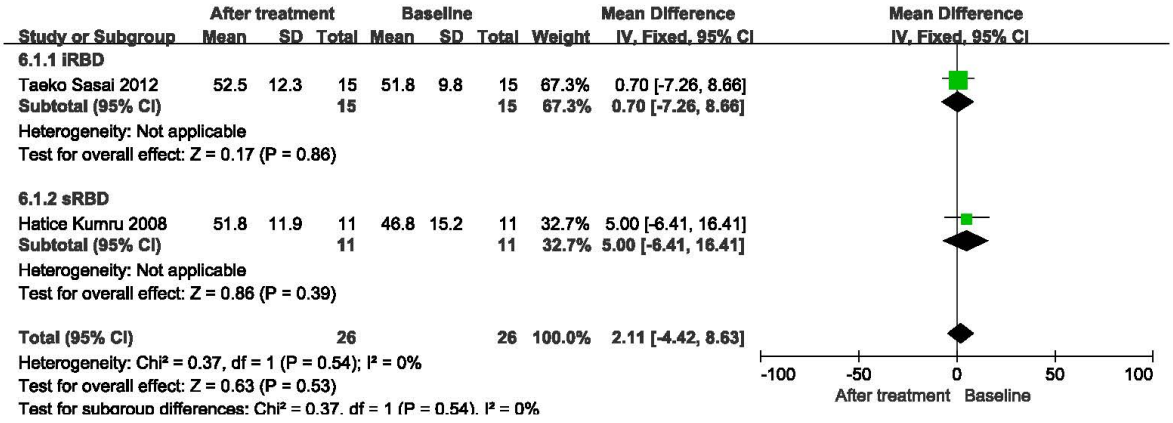
Supplementary figure 7: Forest plot of the meta-analysis showing the effect of pramelteon on TST of different types of RBD



Supplementary figure 8: Forest plot of the meta-analysis showing the effect of melatonin on TST of different types of RBD



Supplementary figure 9: Forest plot of the meta-analysis showing the effect of pramipexole on PLMS index of different types of RBD



Supplementary figure 10: Forest plot of the meta-analysis showing the effect of pramipexole on Stage 2 of different types of RBD.

Characteristics of studies

Characteristics of included studies

DIETER KUNZ 2010

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	This is a randomized double-blind experiment
Allocation concealment (selection bias)	Low risk	This is a randomized double-blind experiment
Blinding of participants and personnel (performance bias)	Low risk	The experiment provided two capsules that participants could not distinguish
Blinding of outcome assessment (detection bias)	Low risk	The recorder is unaware of the patient's information
Incomplete outcome data (attrition bias)	Low risk	Described in the paper, it is not enough to affect the effect of the intervention
Selective reporting (reporting bias)	Low risk	Unlikely to happen
Other bias	Unclear risk	Unclear,other bias might occur

Hatice Kumru 2008

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	unclear,not mentioned in the paper.
Allocation concealment (selection bias)	Unclear risk	unclear,not mentioned in the paper.
Blinding of participants and personnel (performance bias)	High risk	The experiment may have broken the blindness.
Blinding of outcome assessment (detection bias)	High risk	The results of our study, though, were based on subjective reports of patients and bed partners.
Incomplete outcome data (attrition bias)	Low risk	According to the article, we found no key data missing.
Selective reporting (reporting bias)	Low risk	Unlikely to happen.
Other bias	Unclear risk	Unclear,other bias might occur.

Jin-Sun Jun 2019

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The experiment was randomized, double-blind, and the group was determined by random numbers generated by the computer.
Allocation concealment (selection bias)	Low risk	The experiment was randomized, double-blind, and the group was determined by random numbers generated by the computer.
Blinding of participants and personnel (performance bias)	Low risk	There was no difference in the appearance of the drugs provided to each group in the experiment.
Blinding of outcome assessment (detection bias)	Unclear risk	unclear,not mentioned in the paper.
Incomplete outcome data (attrition bias)	Low risk	The article describes the participant's withdrawal from the experiment.
Selective reporting (reporting bias)	Low risk	The reported scale parameters are relatively complete.

Other bias	Unclear risk	Lack of sufficient information to determine the risk of bias.

Kenichi Kashihara 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	unclear,not mentioned in the paper.
Allocation concealment (selection bias)	Unclear risk	unclear,not mentioned in the paper.
Blinding of participants and personnel (performance bias)	High risk	The experiment did not use double blindness.
Blinding of outcome assessment (detection bias)	Unclear risk	Lack of sufficient information to determine the risk of bias.
Incomplete outcome data (attrition bias)	Unclear risk	unclear,insufficient information to judge.
Selective reporting (reporting bias)	Low risk	Unlikely to happen.
Other bias	Unclear risk	unclear,Insufficient information to judge.

M L Fantini 2003

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	unclear,not mentioned in the paper.
Allocation concealment (selection bias)	Low risk	The results of this experiment are controlled by the front and back self-control.

Blinding of participants and personnel (performance bias)	Unclear risk	Not mentioned in the paper.
Blinding of outcome assessment (detection bias)	Low risk	Blinding was used to assess the outcome.
Incomplete outcome data (attrition bias)	Low risk	All pre-declared outcomes have been reported in the article.
Selective reporting (reporting bias)	Low risk	All results were reported.
Other bias	Unclear risk	Lack of sufficient information to determine the risk of bias.

Moran Gilat 2019

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	The group was determined by random numbers generated by the computer.
Allocation concealment (selection bias)	Low risk	The group was determined by random numbers generated by the computer.
Blinding of participants and personnel (performance bias)	Low risk	Cover through the same bottle for different drugs.
Blinding of outcome assessment (detection bias)	Low risk	Blinding was used to assess the outcome.
Incomplete outcome data (attrition bias)	Low risk	There were no missing outcome data,and pre-declared outcomes have been reported in the article.
Selective reporting (reporting bias)	Low risk	Unlikely to happen.
Other bias	Unclear risk	Lack of sufficient information to determine the risk of bias.

N Takeuchi 2001

Methods	
Participants	
Interventions	
Outcomes	

Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	This is not a non-randomized controlled experiment and no blinding was used.
Allocation concealment (selection bias)	High risk	This is not a non-randomized controlled experiment and no blinding was used.
Blinding of participants and personnel (performance bias)	High risk	This is not a non-randomized controlled experiment and no blinding was used.
Blinding of outcome assessment (detection bias)	Unclear risk	unclear,not mentioned in the paper.
Incomplete outcome data (attrition bias)	Low risk	No missing data.
Selective reporting (reporting bias)	Low risk	Not mentioned in the paper but unlikely to happen.
Other bias	Unclear risk	unclear not mentioned in the paper.

Raffaele Ferri 2013

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	All RBD patients with at least one subtentorial vascular lesion or at least two vascular supratentorial lesions greater than 0.5 cm were excluded.
Allocation concealment (selection bias)	High risk	All RBD patients with at least one subtentorial vascular lesion or at least two vascular supratentorial lesions greater than 0.5 cm were excluded.
Blinding of participants and personnel (performance bias)	Low risk	our study included 13 patients analyzed, after approximately 2.5 years of therapy with clonazepam, by means of an unsupervised automatic quantitative method that does not require a blind protocol.

Blinding of outcome assessment (detection bias)	Low risk	The algorithm was run blind to the condition of the subject, even though no manual modifications of the parameters is possible.
Incomplete outcome data (attrition bias)	High risk	There are two cases of missing data.
Selective reporting (reporting bias)	Low risk	Unlikely to happen.
Other bias	Unclear risk	Unclear,not mentioned in the paper.

Shirley Xin Li 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	Newly diagnosed idiopathic RBD (iRBD) patients attending the sleep center were consecutively recruited into this study from 2008 to 2011 as part of another longitudinal study on the clinical outcomes of RBD patients.
Allocation concealment (selection bias)	Unclear risk	Unclear,not mentioned in the paper.
Blinding of participants and personnel (performance bias)	High risk	Treatment assignment was non-randomized. Other more effective or adjunctive interventions and formal RCT studies are needed for better clinical management of RBD.
Blinding of outcome assessment (detection bias)	Low risk	The board-certified PSG technologists, who were blind to the clinical status of the subjects, conducted visual quantification of the REMREEA.
Incomplete outcome data (attrition bias)	Low risk	Patients attended their follow-up appointments in the outpatient clinic, with the treating physicians, on a regular basis. The Institutional Ethics Committee approved the study.
Selective reporting (reporting bias)	Low risk	Unlikely to happen.
Other bias	Unclear risk	Lack of sufficient information to determine the risk of bias.

Taeko Sasai 2012

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	unclear,lack of sufficient information to determine the risk of bias.
Allocation concealment (selection bias)	Unclear risk	unclear,lack of sufficient information to determine the risk of bias.
Blinding of participants and personnel (performance bias)	High risk	the study was not a randomed controlled investigation.
Blinding of outcome assessment (detection bias)	Unclear risk	unclear,lack of sufficient information to determine the risk of bias.
Incomplete outcome data (attrition bias)	Unclear risk	unclear,lack of sufficient information to determine the risk of bias.
Selective reporting (reporting bias)	Low risk	Unlikely to happen.
Other bias	Unclear risk	unclear,lack of sufficient information to determine the risk of bias.

Takashi Nomura 2013

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	This is a case report and no blinding is used.

Allocation concealment (selection bias)	High risk	This is a case report and no blinding is used.
Blinding of participants and personnel (performance bias)	High risk	This is a case report and no blinding is used.
Blinding of outcome assessment (detection bias)	High risk	This is a case report and no blinding is used.
Incomplete outcome data (attrition bias)	High risk	Only partial results were reported.
Selective reporting (reporting bias)	Low risk	Unlikely to happen.
Other bias	Unclear risk	Unclear,not mentioned in the paper.

Yan Wang 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	This is not a randomized double-blind experiment.
Allocation concealment (selection bias)	High risk	This is not a randomized double-blind experiment.
Blinding of participants and personnel (performance bias)	High risk	This is not a randomized double-blind experiment.
Blinding of outcome assessment (detection bias)	Low risk	The data were scored by a board certified sleep specialist blinded to the clinical information.
Incomplete outcome data (attrition bias)	Unclear risk	A double-blinded, placebo-controlled trial of rotigotine treatment in RBD patients should be conducted to confirm the results reported here.
Selective reporting (reporting bias)	Low risk	Unlikely to happen.
Other bias	Unclear risk	Unclear,not mentioned in the paper.

Yuichi Esaki 2016

Methods	
Participants	
Interventions	
Outcomes	
Notes	

Risk of bias table

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	This is an open-labeled trial.
Allocation concealment (selection bias)	High risk	This study has some limitations. First, this study was not a randomized controlled investigation.
Blinding of participants and personnel (performance bias)	High risk	This study has some limitations. First, this study was not a randomized controlled investigation.
Blinding of outcome assessment (detection bias)	Low risk	The v-PSG and quantitative RWA scorings were, as a result, made by impartial raters blinded to the dates of acquisition.
Incomplete outcome data (attrition bias)	Unclear risk	This is an open-labeled trial.
Selective reporting (reporting bias)	Low risk	The v-PSG and quantitative RWA scorings were, as a result, made by impartial raters blinded to the dates of acquisition.
Other bias	Unclear risk	Unclear,not mentioned in the paper.

Footnotes